Surya Shreevathsa K. S

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SUMMARY

Graduate engineer with hands-on C/C++ firmware, Linux, and embedded hardware interface work. Experience writing firmware over UART, I2C, SPI, and CAN, designing PCBs, and building telemetry pipelines on Linux. Comfortable working at the software and hardware boundary, from PCB design through firmware and data collection.

SKILLS

Programming: C, C++, Python, Embedded C

Systems & Tools: Linux, Unix commands, Git, GitHub, FreeRTOS

Embedded & Buses: UART, I2C, SPI, CAN, Firmware development, Raspberry Pi, ESP32, Teensy

Electronics: PCB design (KiCad), Circuit design, Digital electronics, Battery management systems

Web (secondary): React, Node.js, Express, MongoDB, Tailwind CSS

Languages: English, Kannada, Hindi

WORK EXPERIENCE

Embedded Systems Engineer - Meen Motors (Bengaluru) Oct 2025 - Present

Previously Embedded Systems Intern, Jun 2025 - Oct 2025

- Designed the electrical powertrain for a retro EV prototype: battery pack, DC-DC converters, relay switching, VESC motor controller, and wiring laid out for safety and thermal limits.
- Designed and fabricated the vehicle ECU PCB in KiCad (Teensy 4.1, MOSFET gate drivers, level shifting, harness connectors), plus steering-wheel HMI boards and battery sensing and charging circuits.
- Wrote Teensy 4.1 ECU firmware in C++ for vehicle access control, lighting, propulsion enable, BLE owner presence, battery state indication, and USB telemetry to in-car systems.
- Built an onboard telemetry pipeline on Raspberry Pi 5 (Linux) that collects live voltage, current, and battery health data and feeds a dashboard display.
- Integrated dual-lav mic path on the ECU board with analog switching into a phone TRRS jack; firmware selects mics from BLE proximity and the vehicle start button.

Freelance Software Engineer - Vara Jan 2026 - Present

- Build and maintain web products for hospitality clients, including a live SaaS booking platform used by multiple properties.
- Ship responsive sites with Tailwind CSS; built Vara Labs (React, Node.js, Express, MongoDB) with Razorpay payments, currently live with 5 properties.

Intern - Hindustan Aeronautics Limited (HAL) Oct 2023 - Nov 2023

- Observed LRU testing workflows and cockpit avionics systems; gained exposure to safety-critical hardware validation processes and SOPs in a defence manufacturing environment.

EDUCATION

B.E. (EEE) - The National Institute of Engineering, Mysuru 2021 - 2025

Class XII - Jawahar Navodaya Vidyalaya, Kodagu 2020 - 2021

PROJECTS

Design of Electric Go-Kart 2025

- Designed a custom 17s13p lithium-ion battery pack and the high and low voltage architecture, with a safety interlock so the high voltage tractive system can only energise after the low voltage system is live (Accumulator Isolation Relay).
- Integrated safety systems including TSAL, RDS, and AIR-controlled tractive disconnect; validated performance targets in OptimumLap before the prototype was built and driven.

Accident Alert System using Arduino 2024

- Built crash detection with dual-trigger logic (MPU6050 impact plus GPS speed drop to zero) to cut false positives; on confirm, auto-sends an SMS with GPS coordinates to emergency contacts.

Vara Labs 2026

- Full-stack SaaS for property owners (React, Node.js, Express, MongoDB): bookings, pricing, availability, Google OAuth, and Razorpay with webhook verification.

CERTIFICATIONS

Embedded Systems Certification Course: Embedded C, 8051, Arduino and Simulation - ASDC (DIY Guru) 2026

Electric Vehicle Essentials 2: Motors, Electrification and Charging System - ASDC (DIY Guru) 2025

Electric Vehicle Essentials 1: Industry Ecosystem and Battery Technology - ASDC (DIY Guru) 2025

Electronics and Circuit Design for EV Application - ASDC (DIY Guru) 2025

Introduction to the Internet of Things and Embedded System - University of California, Irvine (Coursera) 2024

Sensor and Sensor Circuit - University of Colorado Boulder (Coursera) 2023